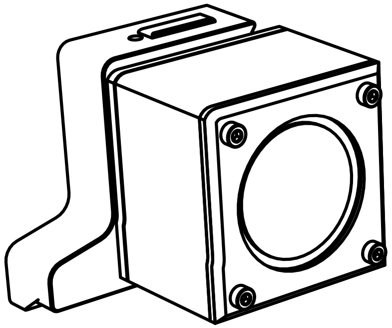
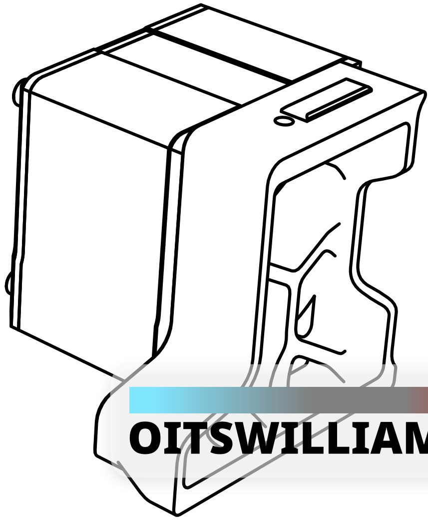
OitswilliamV2

Banger speaker skirts

Assembly guide

REV 2026-06-16





OITSWILLIAM PANG

Hardware

Contents

Before getting started

You are about to read the manual of banger speaker skirts. This manual details about it and assembly of such speakers.

Warning

⚠ Danger: unless if it only runs on DC and only has a DC input, otherwise risk of electric shock

- Even though you can build with this mod from start, if you want to wire, or re-wire, you need to make sure and check that the mains is unplugged.
- Failure doing so which is unplugging the mains power may cause electrocution or even death.
- When in doubt, please don't touch the wires connected to the power supply or SSR.

Other notes

- **There are two parts of model on each side of the skirts**, these being the main part and the modifier. The modifier is used for adding the mesh on the part where it has ventilation or speaker cutouts. Please visit the parts' Readme file on how to prepare these.
- The righthand side of the speaker has another notch that is located at the top, whereas the lefthand side doesn't. This indicates the side and is used for alignment. If the righthand side parts are attached to the lefthand sides, or vice versa, it will be an obvious telltale sign that is incorrect. The box units contain two letters that indicate their sides.
- For their abbreviations, LH means lefthand side and RH means righthand side.

About the banger speakers

This mod adds speakers that sound as loud as it gets, with 30mm radiators to improve bass. It is mounted to Voron 2 150mm (mod). However users can access the CAD model of such model to attach them to other units. Users can install sound extension written by Al Williams to Klippy extra or use aplay command via Klipper Gcode Shell, or just become usable whether the machine becomes a PC.

Contribution

This mod is a part of OitswilliamV2 mod project through GitHub. You can visit, contribute and report new issues here.

Technical support

You can contact Oitswilliam here, or if the issue is not related to OitswilliamV2 mod, you may visit Voron forum or its Discord server.

Guide and legend

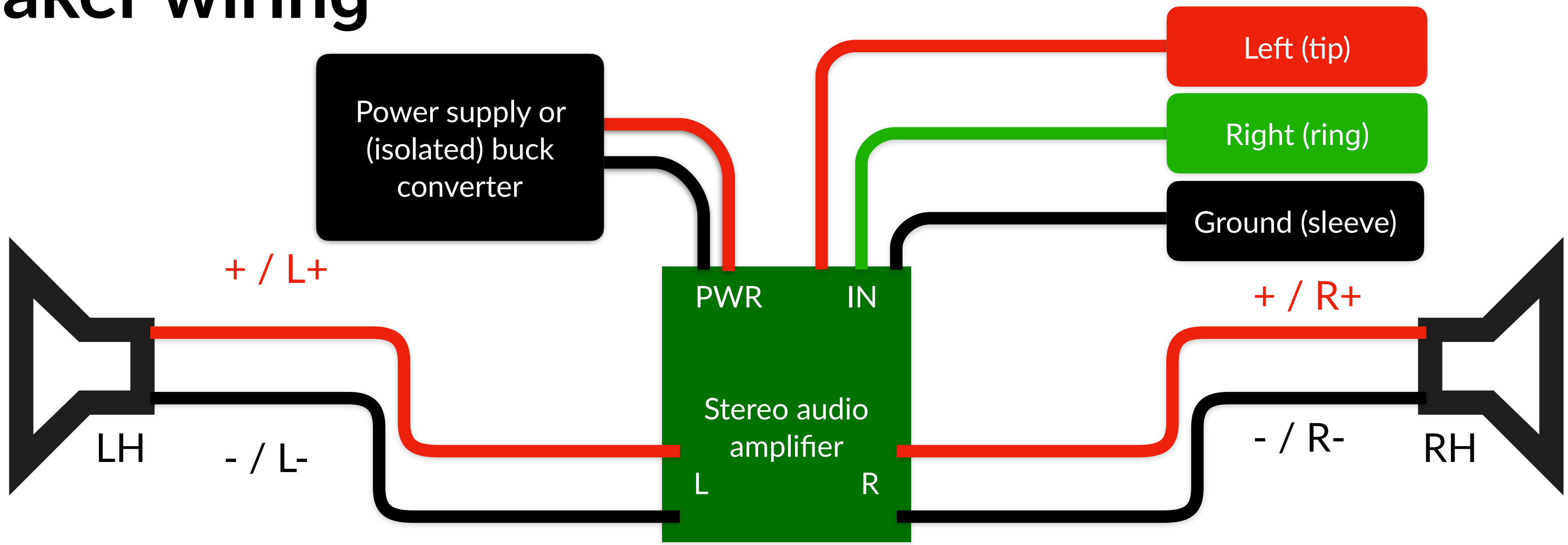
- **Material** - raw material
- **Material** - highlighted material
- **Model** model.stl / model.3mf - printed material, as this part is simple, the printed parts may be color-coded.

Part printing

The parts of this model can be printed in the base color of the printer. **Bold text** indicates the recommended value and *italic text* indicates on what setting / value it needs to set.

FDM / FFF or SLS	Print material: ABS, ASA or PA12, **prohibits PLA.**
Layer height of **0.2mm,** initial layer height can be **0.2mm or higher.**	*Nozzle and extrusion width* must be **forced 0.4mm; if using 0.4mm nozzle**
1.6mm *wall width*, or **4** *wall counts.*	*Infill* must be **at least 40%, except on the infill modifiers**.
At least 4 *solid bottom and top layers*, **except on the infill modifiers.**	The best *print speed* is up to **120mm/s**. The speed might be reduced by the hot end flow rate.
Supports are **recommended for main skirt parts.** Use **tree or organic supports** if available.	*Other settings might be depended on your printer.* *Follow it's warnings for proper handling.*

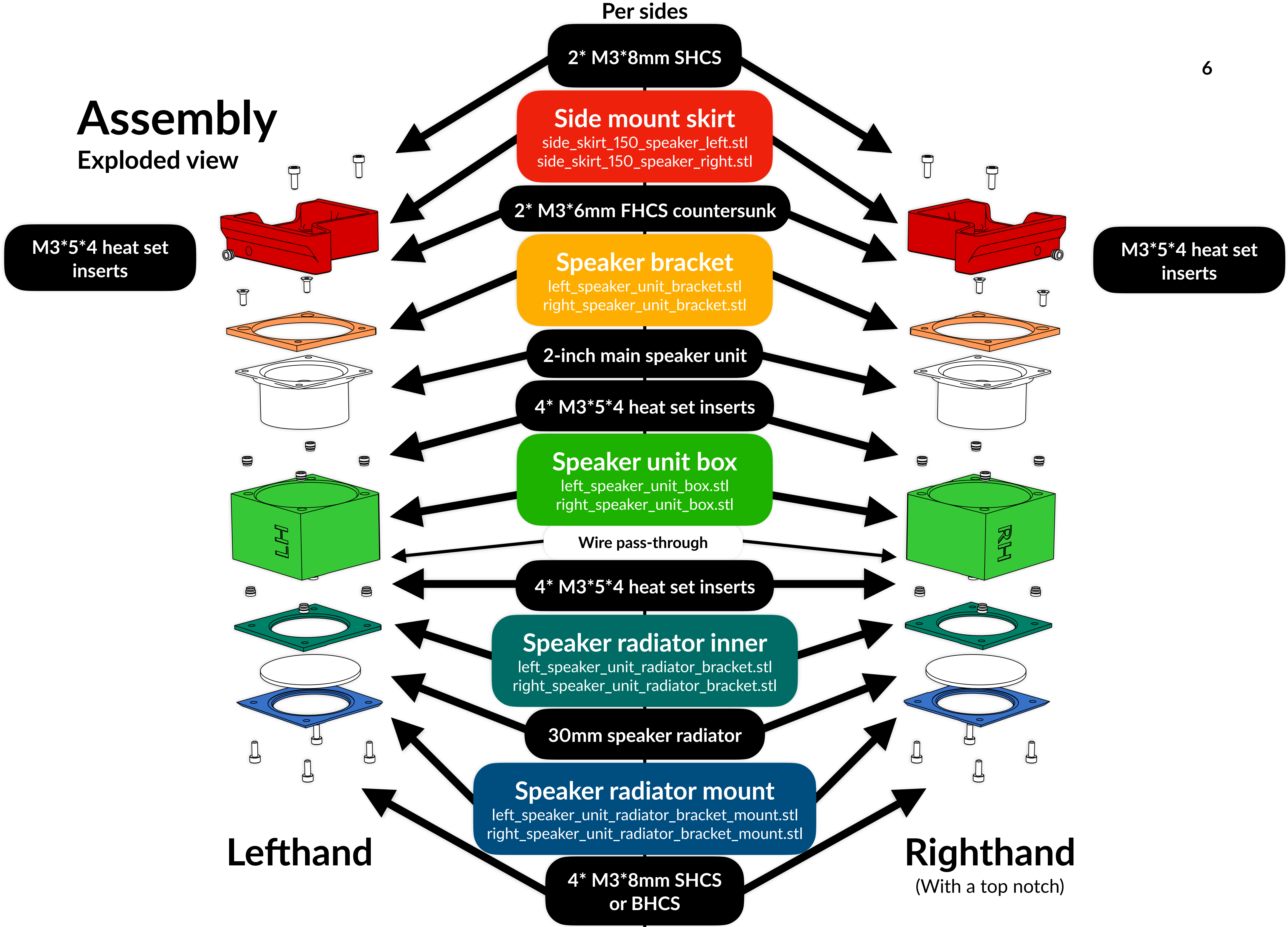
Speaker wiring



- Note that the speaker wire polarity must be matched in order to function correctly, and should not be reversed.
- The power requirements for the amplifiers are different, some of them accept 5V, while others accept 12V or 24V.
- **If the speakers did not have tails of wires, you may make them yourself.** The wire length should be at least 250mm per connection, totaling 1000mm. (On bigger machines the length may be added more)
- **The wiring colors for the analog TRS plug may be different,** because of this, you may use a multimeter to check whether the wire and the jack match.
  - For left (tip), it may use red or blue, for right (ring), it may use green, red or white and for grounding (sleeve) it may be either exposed, or covered with green or black insulation.
- After done wiring and testing, you may unplug them, and prepare them for installation to the printer.

Assembly

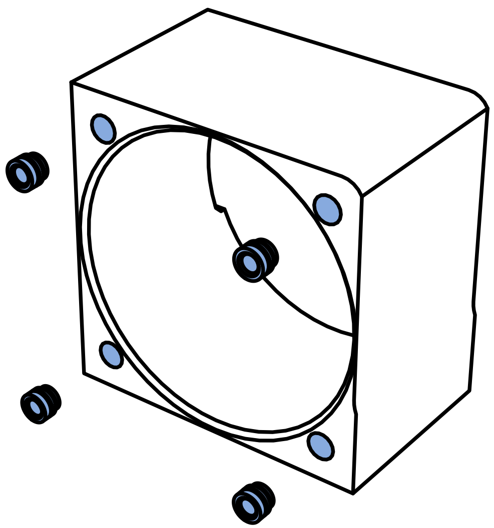
Exploded view



Detailed assembly

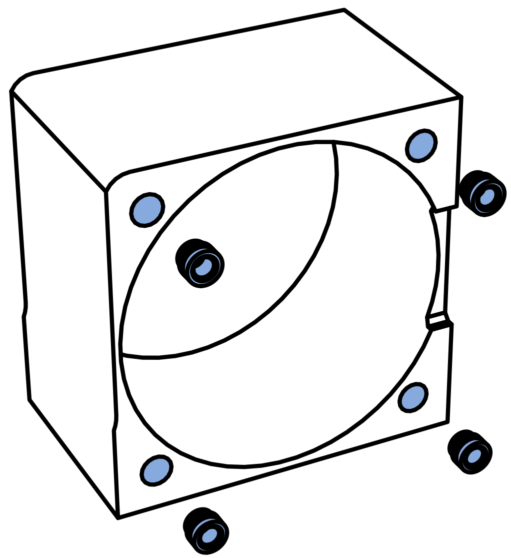
Speaker unit

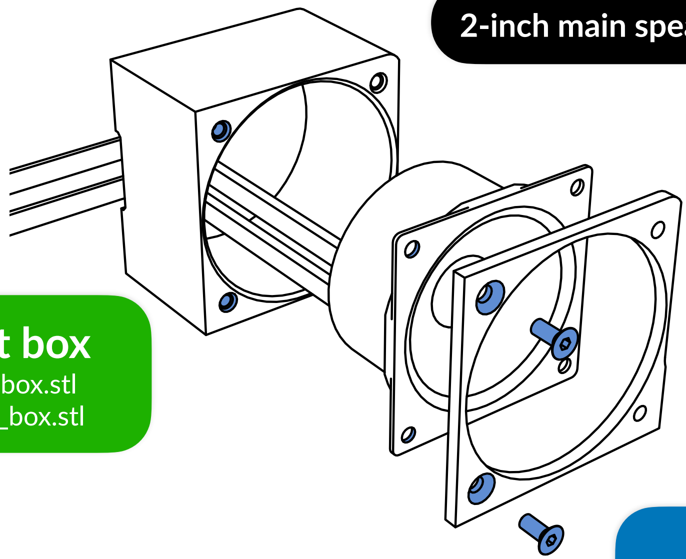
Showing lefthand side only as righthand side is just mirrored and will follow the same pattern, albeit the righthand side has another notch.

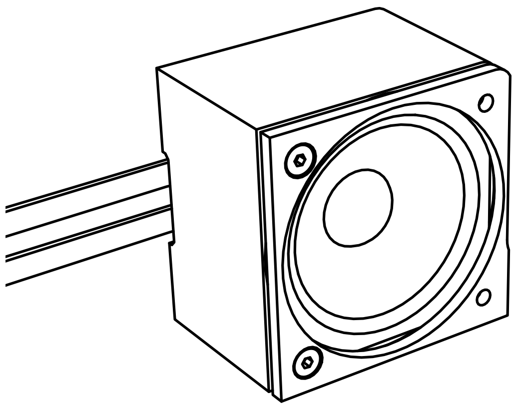


8* M3*5*4 heat set inserts

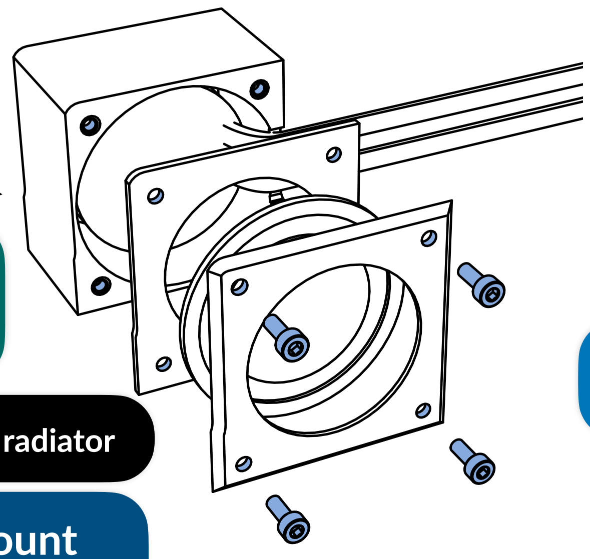
(Per side, totaling 16 on both sides)




Front
2-inch main speaker unit
Speaker bracket
left_speaker_unit_bracket.stl
right_speaker_unit_bracket.stl
The wire should pass through the cutout.
Speaker unit box
left_speaker_unit_box.stl
right_speaker_unit_box.stl
Pay attention to the notch. Please make sure all the parts' notches align.
2* M3*6mm FHCS countersunk
(Per side, totaling 4 on both sides)



Speaker radiator



The wire should pass through the cutout.

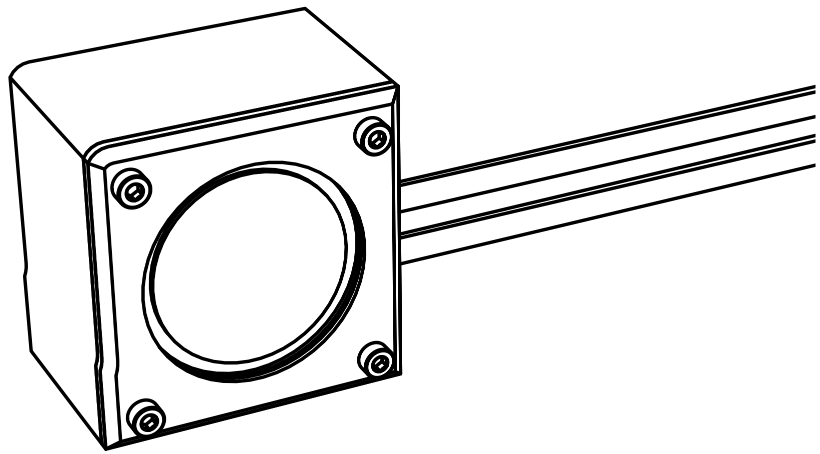
Pay attention to the notch. Please make sure all the parts' notches align.

Speaker radiator inner
left_speaker_unit_radiator_bracket.stl
right_speaker_unit_radiator_bracket.stl

30mm speaker radiator

Speaker radiator mount
left_speaker_unit_radiator_bracket_mount.stl
right_speaker_unit_radiator_bracket_mount.stl

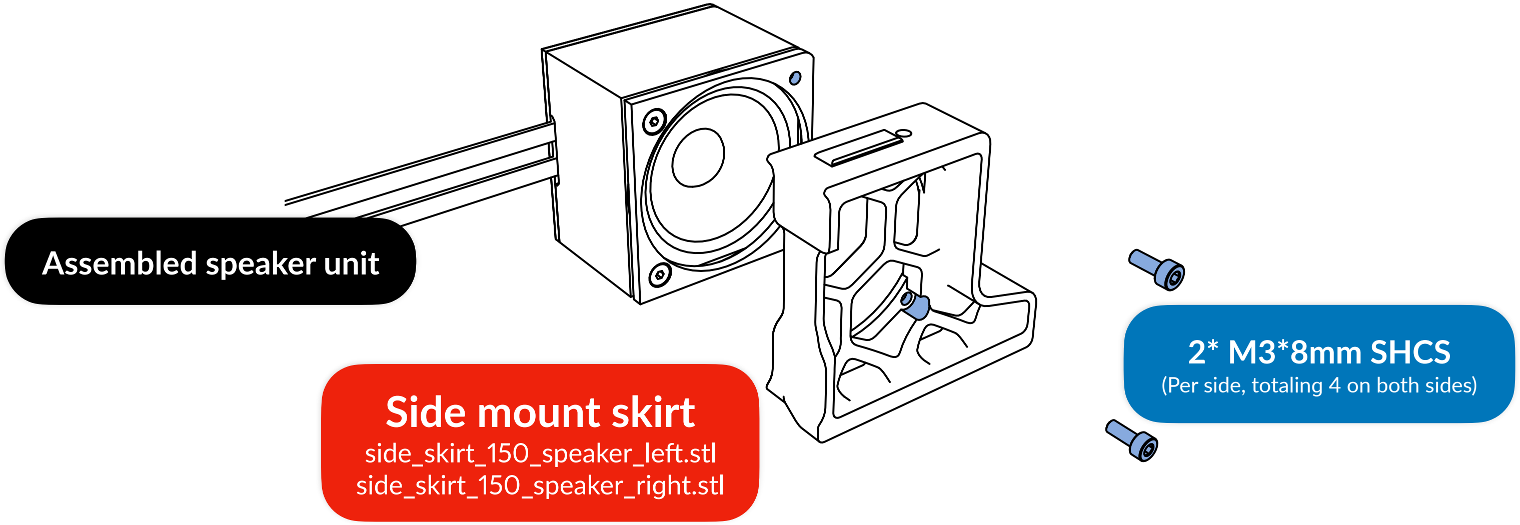
4* M3*8mm SHCS or BHCS
(Per side, totaling 8 on both sides)

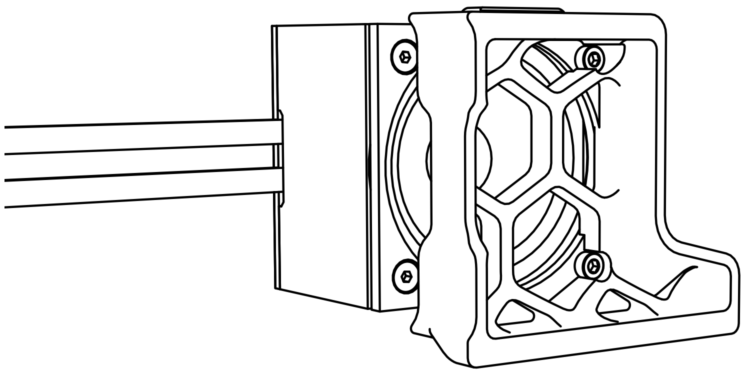


The speaker unit has assembled.

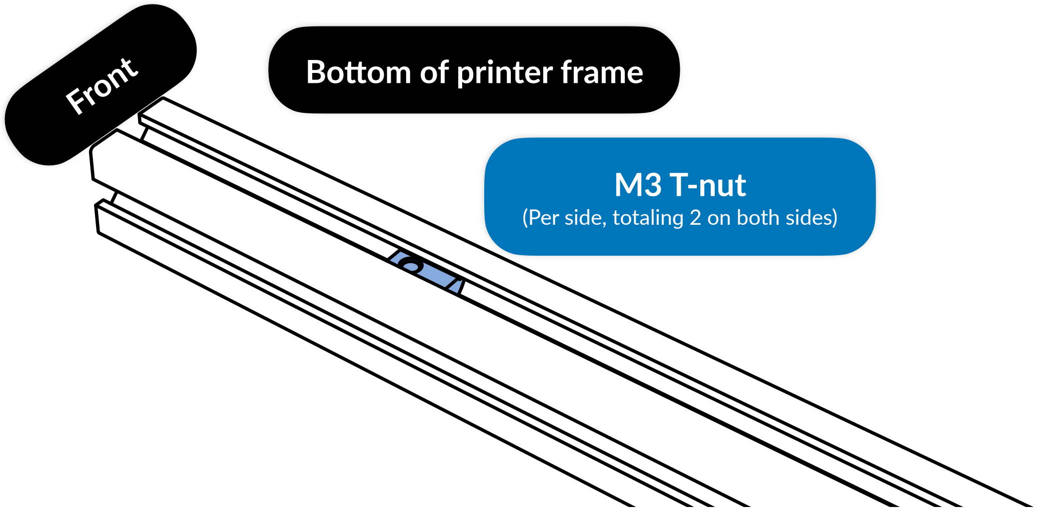
Speaker skirt

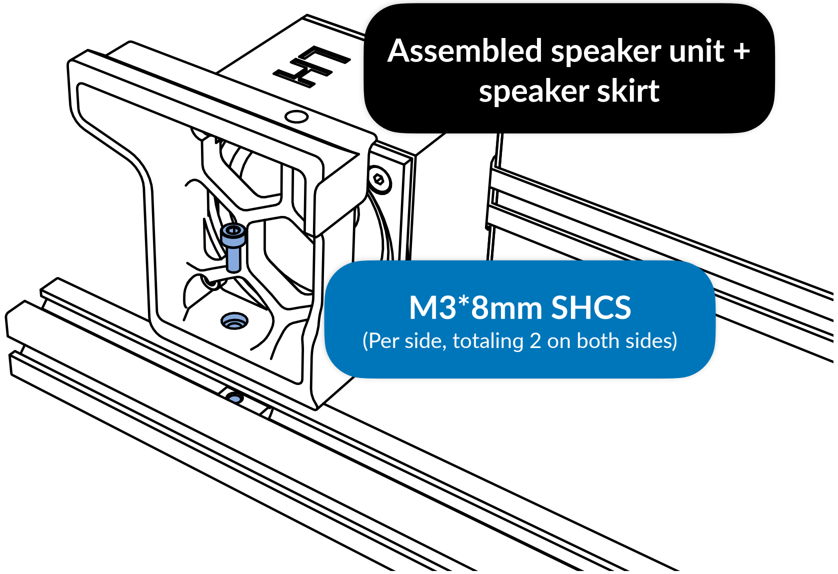
These skirts can be followed by page 226 and 227 of the original V2 assembly manual, with just a quick summary, these skirts can be mounted via M3*8mm screws.

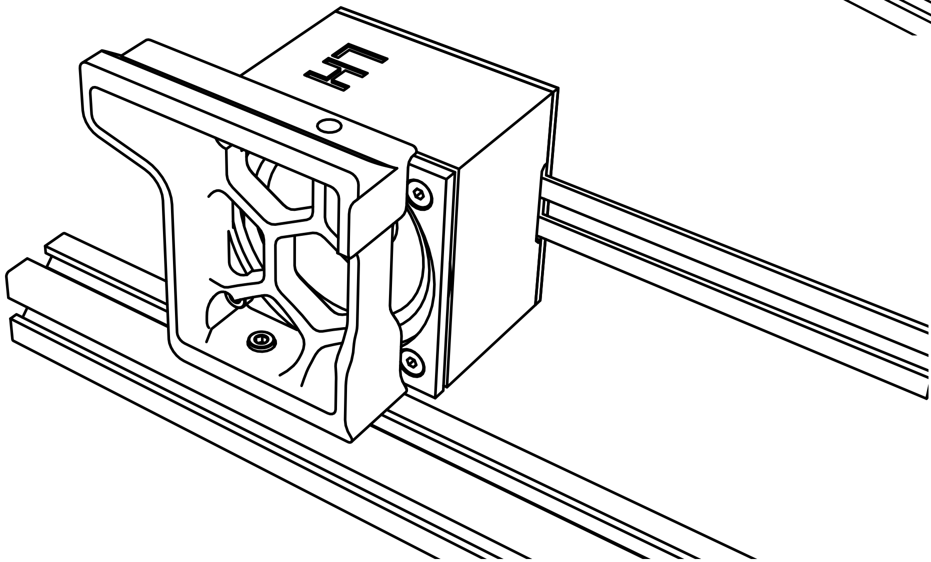




Mounting to frame
Based of page 226 and 227 of original V2 assembly manual.







One side of unit has done. The righthand side may follow the same instructions.

Once all the sides are done, the units can be re-wired, and the deck panel can be reattached.

Software configuration

If the sound has too much treble, which might be, you may see **showing the speakers bass potential**.

Debian / Ubuntu

There is a definite guide from MAGICPHOENIX. If you want to tune ALSA- (alsa-lib) driven audio, you can use AlsaMixer. **Note: the speakers are 10x louder, so it's best to tune down the volume to just 6 to 25%.** You can install libasound2-plugin-equal, and then define it, but this is too difficult and not production-ready.

- You can open AlsaMixer by typing alsamixer at the terminal of the machine.
- In case there are multiple speakers or sound cards, to change the current sound card, press F6 to show the sound card selection menu.
- To adjust the volume, press F5, then use up or down arrow keys to adjust the volume. Since banger speakers are very intrusive, it is generally recommended to set the volume to just 6 to 25%, otherwise there will be a fatal mistake where something yelling emitted.
- You must manually save the changes by typing sudo alsactl store . This requires superuser privilege. Not typing such command will lose changes, and you know what will happen above.
- You can install plugins or even force ALSA to use your preferred speaker / sound card on start-up by creating asound.cfg under /etc/.

If you are using Debian / Ubuntu with desktop environment, you can use EasyEffects to tune and equalize the speakers.

Klipper

Klipper does not have sound playing on its own, but users are able to use Klipper software to allow their machines play audio. There are two ways to make it happen, but the second method is less vulnerable to hacking. If Klipper is in Debian virtual machine, you may also see "**What if Debian is in virtual machine?**".

Uploading audio files to machine

- To place audio files into your machine, the best way to do is to drop & upload the file through lightweight interface (Mainsail / fluidd). First, create a folder called audio just to make sure it is organized, and then drop the audio file you like it to play to such folder.

Klipper Gcode Shell

This method is more popular. There is a video by Stacking Layers and a (previously mentioned) guide from MAGICPHOENIX. Basically, you upload the audio file to the organized directory and then set up the Gcode shell command that allows playing audio via **aplay** and set up the command that executes such shell command.

- Shell command to play sound: aplay /home/pi/printer_data/config/audio/sound.wav # if username is **pi**, using the **modern Klipper file structure** and the **sound** file is **WAV** placed in **audio** subfolder.

Direct implementation

This method is less vulnerable to hacking, if it does not use superuser privilege. There is an article written by Al Williams on Hackaday that puts such implementation to the Klippy extra folder, and users can assign an **aplay** entry and **aplay** Klipper command to play the audio.

Windows

If the enhancements options such as bass are available, you can optionally tune the speakers. To find speaker software enhancements in Windows 11, go to **settings**, and then **system**, then **sound**. After that select **more sound settings**, which will pop up **sounds** window. Select the speaker that powers this and find **enhancements**.

What if Debian is in virtual machine?
If you want sounds played by Klipper or aplay in Debian VM, you need to enable the audio for the virtual machine (you may need to shut it down first), if obviously disabled. On VirtualBox, go to **settings** of Debian VM with Klipper, then select **audio**, and then check **enable audio**, after that, check **enable audio output** under **extended features** section. You may keep the other settings unchanged.

Showing the speakers bass potential

Under Oitswilliam's intensive research, by that I mean 5 minutes, the speakers will show the bass potential (and sound good) on 150Hz at 6dB bass boosted equalizer. You may tune it around such range, around 80 to 200Hz, 3 to 6dB. If the speakers sound good without any tuning, then this is not required. Note that Oitswilliam is not an audiophile expert, please ask the professionals on choosing better hardware around.

Troubleshooting

The speakers are not playing, or is playing through other instruments!
Some software decided to use another instrument instead of the driver that powers banger speakers. You can force using the banger speakers by manually setting.

- On Windows, go to **settings**, and then **system**, then select **sound**. You will see the speakers connected. Select the speaker that power the banger speakers.
- On Debian with GNOME desktop environment / Ubuntu, the procedure is similar, go to **settings**, then **sound**. You will see the speakers connected.
- On Debian **WITHOUT** the desktop environment, you must use ALSA (alsa-lib) CLI. Please check Debian / Linux section for the exact procedure.

Assembly complete

And we are done! If you are satisfied with such product consider tagging @officialbunny350 on Facebook, @oitswilliam on Instagram or @realBunny350 on X (formerly known as Twitter) and share such product with us on Oitswilliam Pang Discord Server.

If you have enquires, problems or issues related to building or using such product, please report it on GitHub.com/Bunny350/OitswilliamV2/issues and we will get on it.

Change log

- 2026-06-16 - Minor adjustment, change Oitswilliam's Instagram handle.
- 2026-06-15 - Update OitswilliamV2 logo and capitalization.
- 2026-03-18 - Add software configuration section about implementing sound output to Klipper and speaker tuning through basic software. Add Instagram link as Oitswilliam came back there.
- 2026-01-24 - Minor adjustments, re-sort the beginning of the manual, minor adjustment to last steps in page 11. Re-use the current logo on the front page.
- 2026-01-08 - initial release

Manual designed by Oitswilliam Pang.
One-person operated design.
www.oitswilliam.com